

Ablation Results — does cognitive attention + contrastive pretraining help?

Auto-generated by `python -m src.analysis.compare_experiments` from the `outputs/metrics/variety_*.json` files each training run produces. Every number here comes from the held-out **test** split, evaluated with the best-validation checkpoint of that run.

Read the Dataset B correction first. The Dataset B table in the next section was produced on a split in which all 127 source seeds had augmented copies in all three splits, so every test image had a same-seed sibling in training. Those numbers measure memorisation and are **withdrawn**; they are printed only as the historical record. The corrected results are in the final section of this document, and they reverse the conclusion. See `docs/09_PROJECT_STATUS_REPORT.md` section 6.9.

Dataset A — corn_3_classes (3 classes)

Experiment	Test accuracy	F1 (macro)	Precision (macro)	Recall (macro)
Baseline (no attention, no contrastive)	99.35%	0.9936	0.9937	0.9936
+ Cognitive attention	100.00%	1.0000	1.0000	1.0000
+ Contrastive pretraining	100.00%	1.0000	1.0000	1.0000
+ Both (proposed)	100.00%	1.0000	1.0000	1.0000

Change vs baseline:

- + **Cognitive attention**: +0.0064 macro-F1 vs baseline
- + **Contrastive pretraining**: +0.0064 macro-F1 vs baseline
- + **Both (proposed)**: +0.0064 macro-F1 vs baseline

3 variants tie at macro-F1 1.0000: + Cognitive attention, + Contrastive pretraining, + Both (proposed). Spread across all trained variants: 0.0064 macro-F1.

Interpretation caveat: the spread between variants is under 0.01 macro-F1, which is within run-to-run noise for a dataset this size. On this dataset the task is close to saturated (see the literature survey — prior work reports 99-100% here), so it cannot discriminate between the architectures. A harder or noisier dataset would be needed to demonstrate a real effect.

Dataset B — maize_data (3 classes) — WITHDRAWN, see correction below

Experiment	Test accuracy	F1 (macro)	Precision (macro)	Recall (macro)
Baseline (no attention, no contrastive)	99.89%	0.9988	0.9988	0.9988
+ Cognitive attention	99.89%	0.9988	0.9988	0.9988
+ Contrastive pretraining	99.96%	0.9996	0.9996	0.9996
+ Both (proposed)	99.92%	0.9992	0.9991	0.9993

Change vs baseline:

- + **Cognitive attention**: +0.0000 macro-F1 vs baseline
- + **Contrastive pretraining**: +0.0008 macro-F1 vs baseline
- + **Both (proposed)**: +0.0004 macro-F1 vs baseline

Best on this dataset: + **Contrastive pretraining** (macro-F1 0.9996). Spread across all trained variants: 0.0008 macro-F1.

Dataset B — CORRECTED, group-aware split

Each of the 127 source seeds appears in exactly one split (89 train / 19 val / 19 test), and the contrastive encoder was pretrained on the training split alone.

Experiment	Test accuracy	F1 (macro)	Seed-level accuracy	n seeds
Baseline (no attention, no contrastive)	85.35%	0.8297	84.21%	19
+ Cognitive attention	85.31%	0.8262	84.21%	19
+ Contrastive pretraining	89.06%	0.8774	94.74%	19
+ Both (proposed)	87.97%	0.8565	84.21%	19

Significance

The 19 test seeds carry ~2,669 images between them, so images are not independent samples. Each seed is scored once (the fraction of its images classified correctly) and variants are compared by paired Wilcoxon signed-rank, Holm-corrected for the six comparisons.

Comparison	Δ per-seed	better/worse/tied	p	verdict
attention_only vs full	+2.63pp	12/2/5	0.0052	significant
baseline vs full	+3.13pp	9/4/6	0.0277	not significant
attention_only vs contrastive_only	+3.00pp	9/4/6	0.3454	not significant
baseline vs contrastive_only	+3.50pp	8/5/6	0.3824	not significant
contrastive_only vs full	-0.37pp	8/4/7	0.6379	not significant
baseline vs attention_only	+0.50pp	7/7/5	0.6832	not significant

The proposed full model significantly outperforms its attention-only ablation. Attention alone remains indistinguishable from baseline, so contrastive pretraining is the component carrying the effect. Note that `contrastive_only` posts the highest mean while failing its own significance test, because its per-seed wins are inconsistent — the characteristic signature of a 19-seed test set. Report this result with n stated, not as a clean win.